

# *flea*

## **NEWS 93** *April 2024*



FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Citations for new flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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## Editorial

### "Flea larvae"

"The discussion is based largely upon the immature stages. This choice is not made because it is thought that they reveal phylogenetic relationships more clearly than do the adults, but because much less is known about the larvae. Their structure makes clear certain relationships previously unsuspected and precludes some derivations once thought possible on the basis of the better known adult structures." - H.E. Hinton (1958)

"The systematics of flea larvae are the least known among larvae of medically important holometabolous insects." - J.R. Kucera (1990)

Dear *Flea News* Reader,

Most fleas are immature. I found over 90% of fleas in a bat roost were immature stages (Bossard 2017). Identification of immature fleas would aid studies of life history, populations, and communities (Haas & Kucera 2004).

In addition, immatures inform phylogenies. Adult and immature phylogeny should in theory be congruent, but in practice they are often in conflict, oft illustrated by other holometabolic insects (e.g. Diptera, Elbel 1988). We might expect "ontogeny recapitulates phylogeny" in flea characters, however, characters may show "caenogenesis", that is, new characters not found in presumed ancestors ("apomorphic" characters).

Holometabolic larvae often have fewer, less complex characters than adults but there are exceptions. Flea larvae sensilla are a possible example: "In holometabolous insects larval sensilla appear to be structurally quite advanced, and more complex than in the adult." (Hallberg & Hansson 1999). Taste is little studied in fleas (King & Gunathunga 2023) although there are studies on food selection.

Some flea larvae reportedly show grouping more distinctively than adults, such as *Ctenophthalmus* that Elbel (1991) placed between Pulicidae and Ceratophyllidae, *Anomiopsyllus*, and with *Rhadinopsylla* possibly a distinct family, similar to the allied group Stenoponiidae.

Elbel also found flea groups he described as having distinct adults but without distinct larvae: the Pulicidae tribes of Archaeopsyllini, Spilopsyllini, and Xenopsyllini group together but are different from Pulicini (contains *Pulex*). Another example is Stenoponiidae whose larvae are not distinct from Hystrichopsyllidae.

Finally, there are flea groups in conflict between larva and adult: *Echidnophaga* larvae resemble Tungidae or Ctenophthalminae not Pulicidae, the bat flea *Sternopsylla* larvae resemble Hystrichopsyllidae not other Ischnopsyllidae, and *Mioctenopsylla* larvae resemble Hystrichopsyllidae not Ceratophyllidae (Elbel 1988, 1996).

Flea larvae description methods are described by Elbel, Smith and Clay 1988, Kucera 1990, Pilgrim 1992, Elbel and Bossard 2007, using light microscope to 800x and Scanning Electron Microscopy (SEM). Larval characters include setae (length, position, pattern, number, morphology), sensilla, antennae papillae and other likely sensory structures, mandibles, and surface denticles. Elbel

thought SEM not worth the trouble, instead opting for DNA as the larvae and their shed exoskeletons (molted exuvia) enables molecular and morphological description (Gurney et al. 2000a, b).

Yours in fleas,  
R.L. Bossard  
Editor, Flea News

Table. To my knowledge, no larvae are described from these families and subfamilies. If you know of any, please send citation to me.

Ancistropsyllidae  
Chimaeropsyllidae  
Coptopsyllidae  
Macropsyllidae  
Malacopsyllidae  
Stephanopsyllidae  
Stivaliidae  
Vermipsyllidae  
Xiphiopsyllidae

or from subfamilies:

Dactylopsyllinae (Ceratophyllidae)

Dinopsyllinae, Doratopsyllinae, Listropsyllinae, Liuopsyllinae (Ctenophthalmidae)

Thaumapsyllinae (Ischnopsyllidae)

Paradoxopsyllinae (Leptopsyllidae)

Moeopsyllinae (Pulicidae)

Parapsyllinae (Rhopalopsyllidae)

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I think these flea taxa contain at least one species with larvae described (incomplete list, some descriptions partial):

Ceratophyllidae

Aetheca  
Callopsylla  
Ceratophyllus  
Citellophilus  
Malareus  
Macrostylophora  
Megabothris  
Mioctenopsylla (larvae resembles Hystrichopsyllidae)  
Monopsyllus  
Nosopsyllus

Opisocrostis  
 Opisodasys  
 Orchopeas leucopus  
 Oropsylla  
 Thrassis  
 Hystrichopsyllidae  
   Anomiopsyllus hiemalis  
   Catallagia  
   Conorhinopsyllus (larva resembles Leptopsyllini)  
   Ctenophthalmus  
   Hystrichopsylla  
   Paratyphloceras  
   Rhadinopsylla  
   Trichopsyllloides  
 Ischnopsyllidae  
   Myodopsylla  
   Nycteridopsylla  
   Sternopsylla  
 Leptopsyllidae  
   Amphipsylla  
   Frontopsylla  
   Leptopsylla segnis  
   Peromyscopsylla  
 Lycopsyllidae  
 Malacopsyllidae  
 Pulicidae  
   Ctenocephalides  
   Cediopsylla  
   Echidnophaga gallinacea (larva resembles Tungini)  
   Euhoplopsyllus  
   Pulex irritans  
   Xenopsylla cheopis  
 Pygiopsyllidae  
 Rhopalopsyllidae  
   Polygenis gwyni  
   Rhopalopsyllus  
 Stenoponiidae  
   Stenoponia  
 Tungidae  
   Tunga monositus  
   Tunga penetrans

### Literature

Barnes, A. M., & Radovsky, F. J. (1969). A new Tunga (Siphonaptera) from the nearctic region with description of all stages. *Journal of Medical Entomology*, 6(1), 19-36. [Tunga monositus]

Bossard, R.L. 2017. Population model for the bat flea *Sternopsylla distincta* (Siphonaptera: Ischnopsyllidae) in temperate caves of the Brazilian free-tailed bat *Tadarida brasiliensis* (Mammalia:

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Chaika, S. Y. (1981). Ultrastructure of the mid-gut of larvae of *Ceratophyllus sciurorum* Schrank (Siphonaptera). *Nauchniye Doklady Vysshei Shkoly, Biologicheskie Nauki*, No. No.1, 61-64.

Morphological studies showed that the epithelial cells of the mid-gut of third-instar larvae of *Ceratophyllus sciurorum* (Schr.) are of the cylindrical type. The apical surface of the cells is modified into microvilli. Very thin filaments of glycocalyx are connected with the outer layer of the membrane of the microvilli. A basal plate, composed of alternating thick and thin fibrilli, underlies the base of the cells. Secretion of enzymes is of merocrine type. Crystals were seen in the cell nuclei; hitherto such crystals had been observed only in adult fleas. A peritrophic membrane was seen in the cavity of the mid-gut of engorged larvae.

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phylogeny of insects. Tested studies for laboratory teaching, 21, 63-77.

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[Add to flea egg bibliography in Flea News 92]

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424.

Wang, D. Q. (1988). A tentative classification of flea larvae (Siphonaptera). *Acta Entomologica Sinica*, 31(3), 326-331.

A tentative classification is given of larvae of fleas known from China to the rank of family, and it is suggested that the egg burster of 1st-instar larvae, the dorsal chaetotaxy of the head, the dorsal and ventral chaetotaxy of the body segments, and the anal comb of 3rd-instar larvae have taxonomic significance. The differences among the genera of each family still remain unknown.

[*Pulex irritans*, *Xenopsylla cheopis*, *Leptopsylla segnis*, *L. lauta*, *Frontopsylla elatoides longa*, *Neopsylla abagaitui*, *N. bidentatiformis*, *N. pleskei orientalis*, *Stenoponia shanghaiensis*, *Ctenophthalmus arvalis*, *Ischnopsyllus indicus*, *Callopsyllus dolabris*, *Ceratophyllus gallinae tribulis*, *Citellophilus tesquorum mongolicus*, *C. trisponus balkhaschensis*, *Monopsyllus anisus*, *Nosopsyllus nicanus*, *N. laeviceps kuzenkovi*, *Oropsylla silantiewi*, *Macrostylophora* sp.]

Williams, B. (1986). Mandibular glands in the endoparasitic larva of *Uropsylla tasmanica* Rothschild (Siphonaptera: Pygiopsyllidae). *International Journal of Insect Morphology and Embryology*, 15(4), 263-268.

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Xiao BoLin, X. B. & Wang, Dun-Qing. (1988) Two flea larvae of *Marmota himalayana*. *Acta Entomologica Sinica*, 1990, Vol. 33, No. 1, 96-98. [*Callopsylla dolabris* and *Oropsylla silantiewi*]

Xiao BoLin, X. B. (1990). Descriptions of larvae of two species of fleas. *Acta Entomologica Sinica*, 1990, Vol. 33, No. 2, 250-253. [*Frontopsylla aspiniformis* and *Amphipsylla qinghaiensis*]

Ye, R. Y., Yu, X., & Wang, D. Q. (1982). Comparative morphology of fifteen species of flea larvae known in China, in addition to four new descriptions and studies on their spiracles. *Acta Entomologica Sinica*, 1982, Vol. 25, No. 2, 209-216.

[*Pulex irritans* L., *Xenopsylla cheopis* (Roths.), *Ctenocephalides felis felis* (Bch.), *Nosopsyllus nicanus* Jordan, *N. laeviceps kuzenkovi* (Yagubyan), *Ceratophyllus anisus* Roths. (*Monopsyllus anisus*), *C. tribulis* Jordan (*gallinae tribulis*), *Neopsylla bidentatiformis* (Wagn.), *N. pleskei* Ioff, *Stenoponia shanghaiensis* Liu & Wu, *Leptopsylla segnis* (Schonh.), *L. lauta* Roths., *Ischnopsyllus indicus* Jordan, *Ctenophthalmus arvalis* Wagn. & Ioff and *Citellophilus trispinus balkhaschensis* (Mikulin).]

Yiming, Q., & Jinhou, H. (1997). Morphological description of the larvae of *Frontopsylla spadix* and their comparison with larvae of three species or subspecies of *Frontopsylla* (Siphonaptera: Leptopsyllidae). *Kun Chong xue bao. Acta Entomologica Sinica*, 40(4), 388-392. [*F. elatoides intermedia*, *F. aspiniformis* and *F. frontalis baibacina*]

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## Announcements

### Lewis World Flea Species List

Available online at Catalogue of Life Checklist, GBIF (Global Biodiversity Information Facility):

Citation: Hastriter, M. W., & Bossard, R. L. (2023). Robert E. Lewis World Species Flea (Siphonaptera) List. In: O. Bánki, Y. Roskov, M. Döring, G. Ower, D. R. Hernández Robles, C. A. Plata Corredor, T. Stjernegaard Jeppesen, A. Örn, L. Vandepitte, D. Hobern, P. Schalk, R. E. DeWalt, M. Keping, J. Miller, T. Orrell, R. Aalbu, J. Abbott, R. Adlard, E. M. Adriaenssens, et al., Catalogue of Life Checklist (Version 2023-06-22). doi: <https://doi.org/10.48580/dfs-48f>

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### Request for Flea Specimens!

I am requesting specimens of Paradoxopsylline genera that are not known from Mongolia, namely:

- *Hopkinsipsylla* from North African jerboa
- *Phaenopsylla* from Calomyscids (Mouse-like hamster)
- *Aconothobius* and *Chinghaipsylla* from pika
- *Calceopsylla* and *Minyctenopsyllus* from zokor
- *Ornithophaga* from woodpecker (Picidae)
- *Typhlomyopsyllus* from Platacanthomyids (Oriental dormice)
- *Xiphiopsylla* from *Tachyoryctes* (African mole-rat) and *Lophuromys* (Brush-furred mouse) or other East African rodent

I would be happy to receive any specimens that could be viable for DNA extraction to sample as much Paradoxopsylline diversity as possible.

Josh Pletcher

jpletche@nmu.edu

### Thesis

Josh Pletcher (Northern Michigan University) is investigating for his master's the phylogeny of Paradoxopsylline fleas and what environmental factors may have driven their diversification (morphology and molecular).

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### **Request for Northern Fowl Mites!**

I am in need of northern fowl mites (a lot of them!) pretty quickly. If anyone has a farm or knows of a farm where there are mites, could you put them in contact with me? I can send information regarding shipping/shipping accounts.

Erika T Machtinger

Associate Professor of Entomology at Penn State University.

University Park, Pennsylvania, United States

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### **M.S. Positions at the University of Texas Rio Grande Valley**

Mays Maestas Vector-borne Disease Ecology Laboratory

UTRGV.edu

Sarah Mays Maestas, Assistant Professor, School of Integrative Biological and Chemical Sciences

sarah.maestas@utrgv.edu

"The Mays Maestas Vector-borne Disease Ecology Laboratory at the University of Texas Rio Grande Valley is searching for candidates for the M.S. program with an interest in vector-borne disease ecology beginning in Fall 2024. The focus of research in the Mays Maestas Lab is on the ecology of vector-borne disease, with a particular focus on tick- and flea-borne zoonoses.

I am particularly interested in host – vector – pathogen interactions and ecological drivers of vector and pathogen presence and abundance. We combine field ecology and the application of molecular tools to improve understanding of the ecology of vector-borne zoonoses and for monitoring and surveillance of vector-borne pathogens.

Current projects include monitoring and surveillance of understudied native tick species and associated pathogens; and investigation of flea species presence, host ranges, and associated pathogens, with a particular focus on vectors of murine typhus. Preference will be given to candidates with wildlife handling experience. Interested individuals should contact Dr. Sarah Mays Maestas (sarah.maestas@utrgv.edu) for more information. Please include a brief description of research interests and experience, along with a current C.V. or resume.

The University of Texas Rio Grande Valley is located in the Lower Rio Grande Valley of south Texas. Positions will be located on the Edinburg, TX campus."

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## **Thank you to the Entomological Society of America (ESA)**

ESA has sponsored Flea News network and archive since 2007. It seems ESA has recently decommissioned its networks and relocated to social media platforms, but the FN archive remains available from the FN editors. We are currently looking for a new internet site for the archive.

- FN editors

## **Parasite Conservation**

### **"New IUCN Species Survival Commission Parasite Specialist Group Launched in 2023"**

"In January 2023, a Parasite Specialist Group was created within the IUCN (International Union for Conservation of Nature) Species Survival Commission. This international group of researchers and practitioners will work to assess the conservation status of parasite biodiversity globally, develop recovery plans and best-practice protocols for parasite conservation, build networks and partnerships for parasite conservation, and communicate the intrinsic and functional importance of parasite species in ecosystems.

The Parasite Specialist Group faces a great challenge. Although approximately half of all animal species are parasitic during at least one life stage, few parasitic species have been assessed for the IUCN Red List or targeted for conservation efforts. Rather than considering all 15 hyperdiverse phyla of understudied and underprotected animal parasites, the Parasite Specialist Group will focus on a subset of parasite biodiversity: metazoan parasites that use vertebrate hosts. This group includes parasitic worms (Cestoda, Trematoda, Acanthocephala, Nematoda), insects (e.g. Phthiraptera, Siphonaptera, Hippoboscoidea, Oestridae, Polyctenidae, Cimicidae) and arachnids (e.g. Ixodida, Spinturnicidae, Trombiculidae). By 2025, the Parasite Specialist Group aims to have assessed the conservation status of at least one representative species of each taxonomic group.

Just as habitat conservation is critical for the conservation of free-living species, host conservation is critical for parasite conservation. Therefore, the Parasite Specialist Groups is seeking new partnerships with vertebrate conservation groups interested in finding ways to conserve parasites along with host species. For example, ex situ vertebrate conservation programmes have resulted in the extinction of some rare parasite species (e.g. the condor louse *Colpocephalum californici*), but they have also sustained some rare and endemic parasite species along with their threatened hosts (e.g. the chewing louse *Ardeicola nippon* on the crested ibis *Nipponia nippon*). An immediate priority for the new Parasite Specialist Group is to develop best practice guidelines for ex situ parasite conservation and work to with interested partners to begin new parasite co-conservation programmes.

**The Parasite Specialist Group invites researchers and practitioners in the fields of parasitology, ecology, veterinary medicine and conservation to reach out with questions, requests for support or ideas for new collaborations."**

– Hopkins, S., & Kwak, M. (2023). *Oryx*, 57(3), 283-283.

– <https://www.iucnparasites.com/get-involved>

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### **Upcoming Meetings**

The 2024 Livestock Insect Workers Conference will be held in Cocoa Beach, Florida, United States, June 23-26, 2024 . To receive deadlines, instructions for registration, paper submission forms, etc. please email Nancy Hinkle (NHinkle@uga.edu) by April 3rd to be added to the mailing list.

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The 4th edition of the Entomology Summer Course will be held in Toulouse, France, from the 8th to 12th of July 2024. This continuing education course will encompass several arthropod groups of medical and veterinary significance (namely sand flies, mosquitoes, *Culicoides* spp., Muscidae, Simuliidae, fleas, bedbugs and ticks). Lectures will generally take place in the morning, while the afternoon will be dedicated to laboratory and field work.

Emilie Bouhsira DVM, MSc, PhD, Dip. EVPC EBVS Specialist in Parasitology Maître de Conférences en Parasitologie - Maladies parasitaires

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The 2024 Society for Vector Ecology meeting will be held in Fort Collins, Colorado, United States, September 15-19, 2024. Titles and abstracts are due April 19th. Other deadlines and information can be found at [SOVE.org/meetings](https://SOVE.org/meetings).

\*\* "Drs. Murrillo and Demard are organizing the Entomological Society of America symposium Innovations in Acarological Research Methodologies: From Lab to Field. The purpose of this symposium is to encourage the exchange of methods and techniques in the field of acarology. The study of mites and ticks often poses unique challenges due to the small size, unique life cycles, and/or morphologies of the taxa. This can create barriers to entry for researchers new to the field of acarology. Here we will showcase researchers across the Entomological Society of America Sections of P-IE, PBT, MUVE, and SysEB with the goal of sharing resources among researchers. This symposium will provide an exchange of ideas, methods, and techniques for researchers at any stage interested in starting their journey with mites or expanding their expertise. Based on the information presented and discussions initiated during the symposium, their goal is to produce an open access special collection featuring publications that will describe the methodology and challenges encountered by researchers. They have already identified a host journal and the organizers will serve as guest editors of the special collection. The sharing of knowledge will promote the continued growth of the field of acarology and empower a new wave of researchers in the field.

Alexandra M. Revynthi, PhD Assistant Professor | Ornamental Entomology and Acarology  
Dept. Entomology and Nematology | University of Florida IFAS

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## Siphonaptera Literature (2023)

(additional to list in fn92)

### New Flea Species

(2024)

Keskin, A. (2024). *Nosopsyllus (Nosopsyllus) spiniformis* n. sp., a new ceratophyllid flea (Siphonaptera, Ceratophyllidae, Ceratophyllinae) from Trkiye. Zootaxa, 5424(3), 389-395.

A new ceratophyllid flea species, *Nosopsyllus (Nosopsyllus) spiniformis* n. sp., is described and illustrated. Specimens were collected from Gnthers ?vole, *Microtus guentheri* (Danford & Alston), *Schidlovskys* vole, *Microtus schidlovskii* Argyropulo, in Adana province, and stone marten, *Martes foina* (Erxleben), in Tokat province. With the new species, the number of *Nosopsyllus* taxa known from Trkiye now tallies 11 (six species and five subspecies).

[*Nosopsyllus (Nosopsyllus) spiniformis* n. sp. Keskin 2024]

(2023)

Eckerlin R.P., Durden L.A.

A NEW SPECIES OF FLEA (SIPHONAPTERA: CTENOPHTHALMIDAE) PARASITIZING VOLES AT HIGH ELEVATIONS IN THE GREAT SMOKY MOUNTAINS OF TENNESSEE AND NORTH CAROLINA.

(2023) The Journal of parasitology, 109 (2), pp. 107 - 112  
DOI: 10.1645/22-85

[abstract fn92]

[ *Catallagia appalachiensis* Eckerlin & Durden 2023 ]

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Hastriter M.W.

Elevation of *Spilopsyllus inaequalis interrupta* (Jordan, 1925) (Insecta, Siphonaptera) to Full Specific Status Based on Morphology and Geographical Distribution.

(2023) Annals of Carnegie Museum, 88 (1), pp. 1 - 12  
DOI: 10.2992/007.088.0101

[abstract fn92]

[ *Spilopsyllus interruptus* (Jordan, 1925) ]

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Li, M. S. (2023). *Waynella* nom. nov. (Siphonaptera: Ceratophyllidae), a new replacement name for *Rowleyella* Lewis, 1971, a junior homonym of *Rowleyella* Weller, 1911 (Brachiopoda: Meristidae). *Systematic Parasitology*, 100(6), 631-632.

*Rowleyella* Lewis, 1971, established for a genus of fleas, was found to be a junior homonym of the brachiopod genus *Rowleyella* Weller, 1911 and is therefore replaced by *Waynella* nom. nov., a substitute that preserves the meaning of the original name. The four previously included nominal species, viz., *Rowleyella arborea* Lewis, 1971 (type-species), *R. gongshanensis* Gong and Duan, 1989, *R. nujiangensis* Lin and Xie, 1990 and *R. fugongensis* Gong and Li, 2020, are accordingly now to be called *Waynella arborea* (Lewis, 1971) n. comb., *W. gongshanensis* (Gong and Duan, 1989) n. comb., *W. nujiangensis* (Lin and Xie, 1990) n. comb. and *W. fugongensis* (Gong and Li, 2020) n. comb., respectively.

### Conservation

Clarke, D. A., & McGeoch, M. A. (2023). Invasive alien insects represent a clear but variable threat to biodiversity. *Current Research in Insect Science*, 4, 100065.

Demetriou, J., Radea, C., Peyton, J. M., Groom, Q., Roques, A., Rabitsch, W., ... & Martinou, A. F. (2023). The Alien to Cyprus Entomofauna (ACE) database: a review of the current status of alien insects (Arthropoda, Insecta) including an updated species checklist, discussion on impacts and recommendations for informing management. *NeoBiota*, 83, 11-42.

Hopkins, S., & Kwak, M. (2023). New IUCN Species Survival Commission Parasite Specialist Group launched in 2023. *Oryx*, 57(3), 283-283. See abstract, in Announcements this FN.

Lymbery, A. J., & Smit, N. J. (2023). Conservation of parasites: a primer. *International Journal for Parasitology: Parasites and Wildlife*.

Nell, C. S., Pratt, R., Burger, J., Preston, K. L., Treseder, K. K., Kamada, D., ... & Mooney, K. A. (2023). Consequences of arthropod community structure for an at-risk insectivorous bird. *PloS one*, 18(2), e0281081.

"Siphonaptera detected in 7.14% of samples"

Rubio-Godoy, M., & de León, G. P. P. (2023). Equal rights for parasites: Windsor 1995, revisited after ecological parasitology has come of age. *Biological Conservation*, 284, 110174.

Lack of data to properly model key biological characteristics like abundance or distribution, seriously hinder comprehension of biodiversity patterns and their drivers — even for historically well-studied and charismatic organisms. Knowledge shortfalls are more extreme in the case of parasites, which are mostly part of the “hidden biodiversity” and generally have a bad reputation. However, being an extremely large portion of biodiversity and a fundamental part of ecosystems, failure to consider parasites is itself an information shortfall for conservation, and can result in important consequences not only for planning strategies to conserve biological diversity, but also to understand fully the patterns and processes that shape ecosystems. Parasites exhibit a marked duality: while some species are pathogenic and do compromise their host's survival (and human interests), the majority exert apparently negligible negative effects on their hosts and could be considered non-pathogenic; indeed, parasites are beneficial as they provide important ecosystemic functions and services. Using a

comprehensive dataset of fish host-helminth parasites in Mexico as a case study, we review knowledge shortfalls relating to parasites, exemplifying their impacts — both positive and negative. A holistic appreciation of parasites should consider both their (many) beneficial and (a few, proven) detrimental impacts, something which could be operationally incorporated into conservation assessments and strategies (e.g., IUCN Green and Red Lists, respectively). Outreach and education are key to modify negative perceptions towards parasites, and to induce recognition of their importance for ecosystemic function and structure.

Ruchin, A. (2023). Mordovia State Nature Reserve as a hotspot of invertebrate biodiversity (European Russia). In E3S Web of Conferences (Vol. 376, p. 02033). EDP Sciences.

### **Taxonomy**

Lareschi M., M. Urdapilleta, AG Autino. 2023. Cap. 43. Siphonaptera. En: Claps, L.E., S.A. Roig-Juñent & J.J. Morrone (Directores). Biodiversidad de Artrópodos Argentinos ISBN 978-950-554-907-8. Volumen 5. volumen 5. Pags. 448-460. Editorial INSUE UNT, San Miguel de Tucumán, Argentina (<https://csnat.unt.edu.ar/investigacion/institutos/insue/publicaciones/libros>).

Maciá, A., Lareschi, M. y Micieli, M.V. 2023. Capítulo 12. Siphonaptera. En "Entomología Médica Y Veterinaria. Biología Y Sistemática De Artrópodos De Interés Médico Y Veterinario En Argentina, Coord. Micieli M. V., Maciá A. & G. Spinelli. Colección Digital Libros de Cátedra 2022, UNLP. <http://sedici.unlp.edu.ar/handle/10915/155007> <https://doi.org/10.35537/10915/155007>

Zurita, A., Lareschi, M. & C. Cutillas. 2023. New insights in the Malacopsylloidea superfamily taxonomy (Siphonaptera). Morphological, molecular and phylogenetic characterization of Phthiropsylla agenoris (Malacopsyllidae) and Polygenis (Polygenis) rimatus (Rhopalopsyllidae). Diversity15(2), 308; <https://doi.org/10.3390/d15020308>

Urdapilleta, M; Lamattina, D; Burgos, E.; Salomón, O.D. & M Lareschi. 2023 Specificity of opossum fleas along an anthropogenic gradient in the Paranaense Rainforest. Zootaxa 10.11646/ZOOTAXA.5264.4.7

Colombo, V.C., Lareschi, M., Monje L.D., Antoniazzi L.R., Morand S. and Beldomenico P.M. 2023 Ecological factors shaping the ectoparasite community assembly of the Azara's Grass Mouse, Akodon azarae (Rodentia: Cricetidae). Parasitology Research <https://doi.org/10.1007/s00436-023-07901-8>

Bossard, R., Lareschi, M, Urdapilleta M., Cutillas C. & A. Zurita. 2023. Flea (Insecta: Siphonaptera) Family Diversity. y. Diversity 2023, 15, 1096. <https://doi.org/10.3390/d15101096>

de la Sancha NU, O.D. Salvioni, C. Vega Gómez & M. Lareschi. 2024. Parasitic laelapid mites (Mesostigmata) on rodents (Cricetidae, Sigmodontinae) from the Interior Atlantic Forest of eastern Paraguay. Sección especial. Los roedores como hospedadores de parásitos. Mastozoología Neotropical 31(1)

Murga Moreno C., D. Ruiz-Pérez, J. Rojas-Moncada, P. Ortiz, L. Sánchez, S. Torrel-Pajares, M. Lareschi. Parásitos del venado de cola blanca (Odocoileus virginianus peruvianus Gray, 1874) en Cajamarca, Perú. Revista Peruana de Biología <https://doi.org/10.15381/rpb.v30i4.26580>

Linardi, P. M. Psocodea. Phthiraptera. In Insetos do Brasil. Diversidade e Taxonomia; Rafael, J.A.;

Melo, G.A.R.; de Carvalho, C.J.B.; Cesari, S.A.; Constantino, R.C., Eds.; Editora INPA: Manaus, Brasil, 2024; pp. 471-483.

Linardi, P. M. Siphonaptera. In *Insetos do Brasil. Diversidade e Taxonomia*; Rafael, J.A.; Melo, G.A.R.; de Carvalho, C.J.B.; Cesari, S.A.; Constantino, R.C., Eds.; Editora INPA: Manaus, Brasil, 2024; pp. 773-782. <https://doi.org/10.3390/d15111153>.

Linardi, P. M. (2023). Fleas as Useful Tools for Science. *Diversity*, 15(11), 1153.

Adult fleas are blood-feeding insects that exclusively infest mammals, acting as parasites and disease vectors. Although certain species exclusively inhabit nests, others are commonly found on the bodies of mammals. Immature stages develop in the soil, inside or near the nests of their respective hosts, making them susceptible to environmental alterations. On hosts, flea infestations are usually defined by abundance, prevalence, and diversity, varying according to host age, sex, size, behavior, habitat, and climate. However, in spite of their vast parasitological importance, fleas have only occasionally been used in applied research. This review focuses especially on the use of mammal fleas as tools or indicators in solving biological, epidemiological, ecological, and phylogenetic questions, and raises new perspectives for future studies.

[complementary to Bossard, R. L., Lareschi, M., Urdapilleta, M., Cutillas, C., & Zurita, A. (2023). Flea (Insecta: Siphonaptera) Family Diversity. *Diversity*, 15(10), 1096. abstract fn92]

Benyahia, H., Parola, P., & Almeras, L. (2023). Evolution of MALDI-TOF MS Profiles from Lice and Fleas Preserved in Alcohol over Time. *Insects*, 14(10), 825.

Chesters, D., Ferrari, R. R., Lin, X., Orr, M. C., Staab, M., & Zhu, C. D. (2023). Launching insectphylo.org; a new hub facilitating construction and use of synthesis molecular phylogenies of insects. *Molecular Ecology Resources*, 23(7), 1556-1573.

The Holy Grail of an Insect Tree of Life can only be ‘discovered’ through extensive collaboration among taxon specialists, phylogeneticists and centralized frameworks such as Open Tree of Life, but insufficient effort from stakeholders has so far hampered this promising approach. The resultant unavailability of synthesis phylogenies is an unfortunate situation given the numerous practical usages of phylogenies in the near term and against the backdrop of the ongoing biodiversity crisis. To resolve this issue, we establish a new online hub that centralizes the collation of relevant phylogenetic data and provides the resultant synthesis molecular phylogenies. This is achieved through key developments in a proposed pipeline for the construction of a species-level insect phylogeny. The functionality of the framework is demonstrated through the construction of a highly supported, species-comprehensive phylogeny of Diptera, built from integrated omics data, COI DNA barcodes, and a compiled database of over 100 standardized, published Diptera phylogenies. Machine-readable forms of the phylogeny (and subsets thereof) are publicly available at [insectphylo.org](https://insectphylo.org), a new public repository for species-comprehensive phylogenies for biological research.

Crowley, L., Allen, H., Barnes, I., Boyes, D., Broad, G. R., Fletcher, C., ... & Sivess, L. (2023). A sampling strategy for genome sequencing the British terrestrial arthropod fauna. *Wellcome Open Research*, 8.

The Darwin Tree of Life (DToL) project aims to sequence and assemble high-quality genomes from all eukaryote species in Britain and Ireland, with the first phase of the project concentrating on



family-level coverage plus species of particular ecological, biomedical or evolutionary interest. We summarise the processes involved in (1) assessing the UK arthropod fauna and the status of individual species on UK lists; (2) prioritising and collecting species for initial genome sequencing; (3) handling methods to ensure that high-quality genomic DNA is preserved; and (4) compiling standard operating procedures for processing specimens for genome sequencing, identification verification and voucher specimen curation. We briefly explore some lessons learned from the pilot phase of DToL and the impact of the Covid-19 pandemic.

Garcia-Rosello, E., Gonzalez-Dacosta, J., Guisande, C., & Lobo, J. M. (2023). GBIF falls short of providing a representative picture of the global distribution of insects. *Systematic Entomology*, 48(4), 489-497.

The Global Biodiversity Information Facility (GBIF) is the largest databank on primary biodiversity data. We examined the completeness and geographical biases for all insect data on GBIF to determine its representativeness. Our results demonstrate that GBIF is far from providing a reliable representation about the global distribution of insects. Despite the growing number of records during the last years, few spatial units are well-surveyed. At coarse resolutions, 34% of the world terrestrial cells lack data and barely 0.5% have completeness values above 90%. Insects are crucial in many ecological functions, and their alarming decline makes it more pressing to have a representative sample to improve our predictive capacity. However, the dynamic nature of species distributions and the strength of anthropogenic forces call for immediate conservation decisions that cannot wait for the empirical data on the identity and distribution of insects.

Hollier, J. (2023). An annotated list of the flea taxa (Insecta: Siphonaptera) described by Fritz Peus, with an account of the holdings of the Peus collection of the Muséum d'histoire naturelle de Genève. *Revue suisse de Zoologie*, 130(2), 127-141.

Medvedev, S. G. (2023). Flea Morphotypes (Insecta, Siphonaptera). *Entomological Review*, 103(5), 506-523.

Muñoz, M. G., Montoya, A., Olmeda, A. S., de Juan, L., Azami, I., Barrena, J. P., ... & Meana, A. (2023). PARASITES IN MOTION: TEACHING VIDEO LIBRARY. In *INTED2023 Proceedings* (pp. 1644-1650). IATED.

Nicholls, S. (2023). *Alien Worlds: How Insects Conquered the Earth, and Why Their Fate Will Determine Our Future*. Princeton University Press.

Sahayaraj, K., & Hassan, E. (2023). Identification of Various Insect Predators. In *Worldwide Predatory Insects in Agroecosystems* (pp. 71-94). Singapore: Springer Nature Singapore.

Sullivan, K. A., Tucker, E. M., Dowdy, N. J., Allen, J. M., Barve, V., Boone, J. H., ... & Zaspel, J. M. (2023). Building a community-based taxonomic resource for digitization of parasites and their hosts. *Insect Systematics and Diversity*, 7(6), 1.

Classification of the biological diversity on Earth is foundational to all areas of research within the natural sciences. Reliable biological nomenclatural and taxonomic systems facilitate efficient access to information about organisms and their names over time. However, broadly sharing, accessing, delivering, and updating these resources remains a persistent problem. This barrier has been acknowledged by the biodiversity data sharing community, yet concrete efforts to standardize and

continually update taxonomic names in a sustainable way remain limited. High diversity groups such as arthropods are especially challenging as available specimen data per number of species is substantially lower than vertebrate or plant groups. The Terrestrial Parasite Tracker Thematic Collections Network project developed a workflow for gathering expert-verified taxonomic names across all available sources, aligning those sources, and publishing a single resource that provides a model for future endeavors to standardize digital specimen identification data. The process involved gathering expert-verified nomenclature lists representing the full taxonomic scope of terrestrial arthropod parasites, documenting issues experienced, and finding potential solutions for reconciliation of taxonomic resources against large data publishers. Although discordance between our expert resources and the Global Biodiversity Information Facility are relatively low, the impact across all taxa affects thousands of names that correspond to hundreds of thousands of specimen records. Here, we demonstrate a mechanism for the delivery and continued maintenance of these taxonomic resources, while highlighting the current state of taxon name curation for biodiversity data sharing.

### Hosts and Distribution

Glebskiy, Y., R. Acosta-Gutiérrez, and Z. Cano-Santana. 2022. Effect of urbanization on the opossum *Didelphis virginiana* health and implications for zoonotic diseases. *Journal of Urban Ecology*, 8(1):1-5. <https://doi.org/10.1093/jue/juac015>

Frixione, M., R. Acosta-Gutiérrez, C. Guzman-Cornejo. 2023. Beyond the last frontier: *Echidnophaga gallinacea* (Siphonaptera: Pulicidae) infestation in a Burrowing Owl, a novel southward host in Mexico. *Zootaxa*, 5351(3): 389-391. <https://doi.org/10.11646/zootaxa.5351.3.7>

Carrillo-González, G. V., Z. Cano-Santana, Y. Glebskiy y R. Acosta-Gutiérrez. 2023. Urbanización y ectoparasitismo en los tlacuaches de la Ciudad de México. *Boletín de la Sociedad Científica Mexicana de Ecología*, 3(8): 32-40. <https://scme.mx/boletin-de-la-scme/>

Vital-García, C., Diana Marcela Beristain-Ruiz, Roxana Acosta, Cinthia Ibeth Prieto Marta, Ana Bertha Gatica Colima, J. F. Aristazabal, Azucena Valdez-Rubio, Carlos Arturo Escudero-Fragosso, Jesús Manuel Martínez Calderas. 2024. Ecological factors shaping ectoparasite communities on Heteromyid rodents at Medanos de Samalayuca. *Parasitology Research*, 123(85) <https://doi.org/10.1007/s00436-023-08098-6>

Acosta, R. J. J. Morrone, C. F. Hernández-Urbina and J. Fernández. 2024. List of the flea species of the state of Chihuahua, Mexico, with new records and a new species of *Strepsylla* Traub, 1950 (Siphonaptera: Ctenophthalmidae: Neopsyllinae: Phalacropsyllini). *Zootaxa*, 5410(2): 239-253. <https://doi.org/10.11646/ZOOTAXA.5410.2.5>

Ahadi, H., Vazirianzadeh, B., Angali, K. A., & Mohammadi, A. (2023). AN INTRODUCTORY STUDY ON FLEAS OF DARREH SHAHR, ILAM PROVINCE, WEST OF IRAN, 2021. *Journal of Experimental Zoology India*, 26(1).

Albers, J. L., Wildhaber, M. L., Struckhoff, M. A., Westrich, D. J., Green, N. S., Poulton, B. C., & Hooper, M. J. (2023). Terrestrial invertebrate diversity and occurrence in restored hardwood forest floodplains, Indiana, United States, June–August 2016 (No. 1168). US Geological Survey.

[Editor's note: Fleas are surprisingly common soil fauna.]

Azarm, A., Saghafipour, A., Yousefi, S., Shahidi, F., & Zahraei-Ramazani, A. R. (2023). Study on Ectoparasites of Free-Ranging Domestic Cats (*Felidae*; *Felis catus*) and Introducing *Trichodectes canis* as a New Record Louse in Tehran Urban Parks, Iran. *Psyche: A Journal of Entomology*, 2023.

Babyesiza, W. S., Mpagi, J., Ssuuna, J., Akoth, S., & Katakweba, A. (2023). Ectoparasite fauna of rodents and shrews with their spatial, temporal, and dispersal along a degradation gradient in Mabira Central Forest Reserve. *Journal of Parasitology Research*, 2023.

Barili, Ö., Tuygun, T., Topçu, E. B. G., & Umur, Ş. (2023). The Parasites of Cats in Türkiye. *Türkiye Parazitolojii Dergisi*, 47(3), 190-199.

Benkacimi, L., Diarra, A. Z., Bompar, J. M., Bérenger, J. M., & Parola, P. (2023). Microorganisms associated with hedgehog arthropods. *Parasites & Vectors*, 16(1), 211.

Bendjeddou, M. L., Khelfaoui, F., Abiadh, A., Mechouk, N., Mihalca, A. D., & Sándor, A. D. (2023). Bat Ectoparasites (Acari, Diptera, Hemiptera, Siphonaptera) in the Grand Maghreb (Algeria, Libya, Mauritania, Morocco and Tunisia): A Literature Review and New Data. *Acta Parasitologica*, 1-15.

Berlov, O. E., & Artemyeva, S. Y. (2023). To the fauna of fleas (Insecta, Siphonaptera) of the reserve “Baikalo-Lensky”. *Parazitologîâ*, (2), 167-171.

The fauna of fleas of the Baikalo-Lensky State Nature Reserve (Siberian Federal District, Irkutsk Region) has been poorly studied. One of the authors previously published information about 7 species collected in the reserve, based on the materials of two field seasons in 1997 and 1998. As part of the inventory of the fauna of ectoparasites of the reserve, during seven field seasons, in 2016-2022, new material was collected. The purpose of the research is to make an inventory of the species composition of fleas that parasitize small mammals. As a result of the work, 384 fleas were caught. The collected fleas belong to 16 species from three families. The most interesting find is the flea *Catallagia striata* Scalon, 1950, which was first encountered in Irkutsk region.

Bourlat, S. J., Koch, M., Kirse, A., Langen, K., Espeland, M., Giebner, H., ... & Fonseca, V. G. (2023). Metabarcoding dietary analysis in the insectivorous bat *Nyctalus leisleri* and implications for conservation. *Biodiversity Data Journal*, 11.

[one unidentified flea species detected in guano]

Carbonara, M., Mendonza-Roldan, J. A., Perles, L., Alfaro-Alarcon, A., Romero, L. M., Murillo, D. B., ... & Otranto, D. (2023). Parasitic fauna of bats from Costa Rica. *International Journal for Parasitology: Parasites and Wildlife*, 20, 63-72.

Carrilho, M., Teixeira, D., Silva, M., Nunes, M., Vieira, M. L., Novo, M. T., ... & Rosalino, L. M. (2023). Prevalence of ectoparasites on small mammals inhabiting an agroforestry system of western Portugal. *Comparative Parasitology*, 90(1), 19-26.

Chalchisa, A., & Bersissa, E. Ectoparasites and their effect on camels (*Camelus dromedarius*) in Ethiopia. *J. Vet. Res. Adv.*, 05(01): 55-69.

da Silva, T. X. D. A., de Almeida, G. P. S., Borges, D. A., Rios, V. E. C., & Correia, T. R. (2023). *Cavia porcellus* (Rodentia: Caviidae) as host for *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) in

artificially infestation. Brazilian Journal of Veterinary Medicine, 45.

The number of guinea pigs is increasing as pet and their domestication necessitates the study of their pathology and emerging diseases. This study aimed to evaluate guinea pigs' capacity to be infested by *Ctenocephalides felis felis* fleas, as it is a common parasite among pets that causes irritation, stress, and transmission of other pathogens. Seventeen guinea pigs were infested with *C. felis felis*. After 48 hours, the animals were combed, and the number of fleas was determined. Guinea pigs had a very low recovery (average of 5%), leading us to conclude that they are not good hosts for this species, despite some literature citing it as an accidental host if infested along with dogs or cats.

Di Benedetto, I. M. D., Oscherov, E. B., & Autino, A. G. (2023). Ectoparasites (Diptera, Hemiptera and Siphonaptera) of bats (Chiroptera) in northeastern Argentina. *Sociedad Chilena de Entomología; Revista Chilena de Entomología*; 49; 4; 11-2023; 773-788.

New information is presented regarding bats and their ectoparasite insects in the Esteros del Iberá ecoregion, Corrientes, Argentina. The aim of this study was to explore the diversity of ectoparasites species found on bats in northeastern Argentina and analyze their taxonomic features, hosts, and distribution. Bats were collected from the Molossidae, Pyllostomidae and Vespertilionidae families along with ectoparasites from the Streblidae, Nycteribiidae (Diptera), Polyctenidae (Hemiptera) and Ischnopsyllidae (Siphonaptera) families. *Aspidoptera falcata* Wenzel, 1976, *Megistopoda theodori* Wenzel, 1966, *Hesperoctenes eumops* Ferris & Usinger, 1939 and *Hesperoctenes fumarius* Westwood, 1874 were added to the ectoparasite fauna for the province of Corrientes, increasing the number of species and their geographic distribution. A rarely studied stage of a larva about to emerge from the posterior end of *Basilia carteri* Scott, 1936 (Nycteribiidae) was recorded. Additionally, 13 parasitological associations were made for the Esteros del Iberá, four of which are new for the province of Corrientes: *B. carteri* with *Myotis riparius* Handley, 1960, *H. fumarius* and *H. giganteus* Ronderos, 1959 with *Molossus rufus* É. Geoffroy Saint-Hilaire, 1806 and *Myodopsylla wolffsohni wolffsohni* Rothschild, 1903 with *M. riparius* Handley, 1960; the latter being also new for Argentina. These results provide host and locality records and new parasite-host associations

[*Myodopsylla wolffsohni wolffsohni* (Rothschild, 1903)]

Dik, B. (2023). Studies on the Determination of Ectoparasites and the Prevalence Rate in Cattle in Izmir, Aydin and Mugla Regions/Izmir, Aydin ve Mugla. *Turkish Journal of Parasitology*, 47(4), 244-249.

Dong, L., Li, Y., Yang, C., Zhu, W., Huang, Y., Lu, S., ... & Yang, J. (2023). Species-level microbiota of ticks and fleas from *Marmota himalayana* in the Qinghai-Tibet Plateau. *Frontiers in Microbiology*, 14, 1188155.

Eckerlin, R. P. (2023). A Collection of Fleas from the District of Stikine of British Columbia, Canada. *Proceedings of the Entomological Society of Washington*, 125(1), 63-67.

Ermolova, N. V., Artyushina, Y. S., Lazarenko, E. V., Bammatov, D. M., Grigoriev, M. P., Klimova, L. I., ... & Halidov, A. H. (2023). Fleas (Siphonaptera) of mammals and birds of the southern part of the Caspian sandy natural plague focus. *Parazitologiya*, (2), 137-150.

Fitzgerald, B. M., Efford, M. G., & Karl, B. J. (2023). The fleas of house mice (*Mus musculus* L.) and

ship rats (*Rattus rattus* L.) in forest of the Orongorongo Valley, New Zealand. *New Zealand Journal of Zoology*, 1-16.

["*Leptopsylla segnis* and *Nosopsyllus fasciatus* were common on mice and rats respectively"]

FRIXIONE, M. G., ACOSTA-GUTIÉRREZ, R., & GUZMÁN-CORNEJO, C. (2023). Beyond the last frontier: *Echidnophaga gallinacea* (Siphonaptera: Pulicidae) infestation in a Burrowing Owl, a novel southward host in Mexico. *Zootaxa*, 5351(3), 389-391.

Galan, M., Bordes, A., Gauthier, P., Kane, M., Niang, Y., Pierre, É., & Granjon, L. (2023). The diet of commensal *Crocidura olivieri* (Soricomorpha: Soricidae): predation on co-existing invasive *Mus musculus* suggested by DNA metabarcoding data. *Mammalia*, 87(4), 326-334.

[*Xenopsylla cheopis*]

Gebrezgiher, G. B., Makundi, R. H., Katakweba, A. A., Belmain, S. R., Lyimo, C. M., & Meheretu, Y. (2023). Arthropod ectoparasites of two rodent species occurring in varied elevations on Tanzania's second highest mountain. *Biology*, 12(3), 394.

Hador, A. B., Ekhnefer, A. M., Bowashia, S. M., & Mohammed, F. F. Identification and morphological characterization of two human and dog fleas species in Al-Wusita-Libya. *Libyan Journal of Basic Sciences (LJBS)* Vol: 21, No:1 , P:38-49, August. 2023 <https://ljbs.omu.edu.ly/> eISSN 2707-6261

Hassan Abdullah, S. H. A. D. A. N. (2023). Ectoparasite Infestation Of Goats (*Capra Hircus*) In Northern Iraq: Prevalence And Species Identification. *Assiut Veterinary Medical Journal*, 69(177), 16-26.

Huang, S., Zhao, M., Luo, X., Bedos, A., Wang, Y., Chocat, M., ... & Liu, W. (2023). Feihu Dong, a new hotspot cave of subterranean biodiversity from China. *Diversity*, 15(8), 902.

Karbowiak, G., Stanko, M., Smahol, K., Werszko, J., & Rychlik, L. (2023). Parasitic Arthropods of Soricinae Shrews in North-Eastern Poland. *Animals* 2023, 13, 2960.

Keskin, A., & Dik, B. (2023). First data on the ectoparasites (ticks, lice and fleas) of the stone marten, *Martes foina* (Erxleben) in Turkey. *Arthropods*, 12(3), 141-147.

Türkiye is a natural land bridge connecting Europe and Asia, and has a rich terrestrial mammalian fauna, with more than 150 species within the seven orders. However, there is limited information on ectoparasites of wild animals, especially carnivores, in Türkiye. In the present study, we provide the first data of ectoparasites of Turkish stone marten, *Martes foina* (Erxleben). A total of 38 ectoparasite specimens were collected from two road-killed stone martens in two different localities in Tokat province. Here, the presence of the genus *Stachiella* Kéler (Insecta: Phthiraptera) in Türkiye was reported for the first time, based on two specimens (1 male and 1 nymph) of *Stachiella retusa* (Burmeister). In addition, three flea (Insecta: siphonaptera), namely *Monopsyllus sciurorum sciurorum* (Schränk) (1 male), *Pulex irritans* L. (1 female), *Chaetopsylla rothschildi* Kohaut (3 males, 8 females) and two tick (Acari: Ixodida) species, namely *Haemaphysalis erinacei* Pavesi (18 males, 4 females), *Ixodes redikorzevi* Olenov (1 female) were collected from stone martens. All ectoparasitic species are new host-associations for stone marten, *M. foina*, in Türkiye.

Kitrytė, N., & Baltrūnaitė, L. (2023). Ectoparasitic mites, ticks (Acari: Trombidiformes, Mesostigmata, Ixodida) and insects (Insecta: Psocodea, Siphonaptera) of ground-dwelling small mammals in the Baltic States. An annotated checklist. *Zootaxa*, 5353(1), 1-46.

Kondratev, E. N., Sazhnev, A. S., Anikin, V. V., & Mironova, A. A. (2023). USING THE HETEROCENTRIC MODEL IN POPULATION-CONSORTIUM ANALYSIS OF THE NEST-DWELLING ARTHROPODS OF THE SAND MARTIN (*RIPARIA RIPARIA* (LINNAEUS, 1758)) IN SARATOV REGION. *Russian Journal of Ecosystem Ecology*, 8(2), 22-31.

Kotti, B. K. (2023). Fleas (Siphonaptera) of rodents in the Transcaucasian highland natural plague focus. *Parazitologîa*, (2), 151-166.

The Transcaucasian highland vole natural plague focus is located within the Javakheti-Armenian (Javakheti and Armenian) Highlands and the Lesser Caucasus. In this region, rodents are hosts for fleas of 42 species from 21 genera. The one third part of flea species are monoxenous, the remaining species are oligoxene or pleioxene parasites. Fleas of 24 species are represented throughout the focus. Others are confined either to the Armenian or the Javakheti Highlands, as well as to the Lesser Caucasus. On the territory of the Transcaucasian focus, the warm period of the year is the most favorable time for the transmission of the plague, since fleas - parasites of the common vole - are characterized by high activity of nutrition and reproduction during this period.

Lee, J. S., Lee, D. J., Lim, J., Lee, H., Kim, J., Choi, M. J., ... & Roh, S. J. (2023). Establishment of an integrated island species inventory for insects (Arthropoda: Insecta) in Korea. *Journal of Asia-Pacific Biodiversity*.

Lewis, M. J. (2023). Preliminary Survey of the Insects of the Ouachita Mountains Biological Station, Polk County, Arkansas. Louisiana State University in Shreveport.

Lignon, J. S., Pinto, D. M., Monteiro, S. G., de Mello, G. T. C., Martins, N. S., Dos Santos, T. S., ... & Bruhn, F. R. P. (2023). *Ctenocephalides felis* (Siphonaptera, Pulicidae) parasitizing White-eared opossum (*Didelphis albiventris*) at Southern Brazil-Case report. *Brazilian Journal of Veterinary Medicine*, 45.

*Didelphis albiventris* is a marsupial with synanthropic habits, whose contact with domestic animals and humans is increasing. They are considered potential disseminators of diseases, present as definitive hosts or reservoirs of various pathogenic agents. The present study aim to report, for the first time, the parasitism by *Ctenocephalides felis* in *D. albiventris* in Rio Grande, Southern Rio Grande do Sul state, Brazil. An adult female White-eared Opossum (*D. albiventris*), killed by being run over, was collected on the roads of the city of Rio Grande, Rio Grande do Sul, Brazil and sent to the laboratory for necropsy. During external inspection, a flea was found, which was stored in 70o alcohol until taxonomic identification, being classified as *C. felis*. This study reports the occurrence of parasitism by *C. felis* in *D. albiventris* for the first time in the location studied, demonstrating that these arthropods circulate between the wild and urban environments, and play an important role in the cycle of zoonotic pathogens among wild fauna, companion animals and humans.

Medvedev, S. G., Verzhutsky, D. B., & Kotti, B. K. (2023). Palaearctic Flea Species of the Genus *Xenopsylla* (Siphonaptera: Pulicidae) Parasitic on Gerbils (*Rhombomys*, *Meriones*) and Their Role in Natural Plague Foci. *Entomological Review*, 103(4), 401-421.

Medvedev, S. G., Verzhutsky, D. B., & Kotti, B. K. (2023). Species of the genus *xenopsylla* (siphonaptera; pulicidae) as vectors of plague infection in its natural foci. *Parazitologîa*, 57(4), 267-309.

Medvedev, S. G., & Stanyukovich, M. K. (2023). Fleas (Siphonaptera) of Small Mammals in the Cape Kartesh Area (the White Sea) and Distribution of These Species in Northwestern European Russia. *Entomological Review*, 103(4), 422-431.

Najera-Cortazar, L. A., Keen, A., Kitching, T., Stokes, D., & Goodman, S. J. (2023). Phylogenetic analyses reveal bat communities in Northwestern Mexico harbor a high diversity of novel cryptic ectoparasite species. *Ecology and Evolution*, 13(2), e9645.

Nikita, Jannatul and Nath, Tilak Chandra and Bhuiyan, Md. Jamal Uddin and Islam, Saiful and Islam, Kazi Mehetazul and Basir, Samiul and Dey, Nilotpal and Bhattacharjee, Tiluttom and Lee, Dongmin, Morpho-Molecular Investigation of Ectoparasitic Infestation of Companion Animals in Sylhet City, Bangladesh. Available at SSRN: <https://ssrn.com/abstract=4550832> or <http://dx.doi.org/10.2139/ssrn.4550832>

Ticks (Ixodida) and Fleas (Siphonaptera) are considered amongst the most important arthropod of public health concern due to their ability to transmit vector-borne pathogens to humans. By sharing a common environment, vector-borne diseases constituted major setbacks to the development of a pet population in Bangladesh. This study aimed to determine companion animal-associated ticks and fleas based on morpho-molecular approaches. Between December 2021 and May 2022, 74 animals (62 cats and 12 dogs) were examined, of which 17 (27.4%) cats and 9 (75.0%) dogs had ectoparasitic infestations, with 35.1% overall prevalence. Morphometrical examination showed the ectoparasites in these animals were *Ctenocephalides* spp. (flea) and *Rhipicephalus* spp. (tick). Genetic analysis using the mitochondrial markers i.e. Cytochrome c oxidase subunit 1 (cox1) revealed the presence of two flea genera i.e., *Ctenocephalides canis*, *Ctenocephalides felis*, and one tick genera *Rhipicephalus sanguineus*. Interviews of animal owners indicate that 35.14% of them had no concern about ectoparasitic infestation or ectoparasites-borne diseases. Our results indicated that fleas and ticks were the most common ectoparasites in companion animals of this area. The zoonotic nature of some ectoparasites can be regarded as a public health alert.

Nizamov, N. S. (2023). Identification of ectoparasitic insects among domestic goats in Bulgaria. *Veterinary World*, 16(4), 728.3

The study demonstrated that the species *L. stenopsis*, *L. africanus*, *B. caprae*, and *P. irritans* were encountered in more than 40% of surveyed farms, situated in 68.75% of regions in Bulgaria. The most intense infestation was by species from the *Linognathus* genus (907 insects), whereas the highest extensity of infestation was registered for *P. irritans* (32.3%). This study detected *P. irritans* as the only flea species.

Nogueira, B. C. F., Cassiano, L. D. A., Martins, T. F., Graciolli, G., Ribon, R., Yamatogi, R. S., & Campos, A. K. (2023). Records of ectoparasites on humans and wildlife in southeastern Brazil. *Archives of Veterinary Science*, 28(1).

Nsengimana, V., Iradukunda, C. S., de Dieu Nsenganeza, J., Mberwa, J. W., & Dekoninck, W. (2023). Soil-litter arthropod communities under pasture land use in southern Rwanda. *Tropical Ecology*, 64(2), 369-379.

Öktem, Ö., & Dik, B. (2023). Studies on the Determination of Ectoparasites and the Prevalence Rate in Cattle in İzmir, Aydın and Muğla Regions. *Türkiye Parazitolojii Dergisi*, 47(4), 244-248.

**Objective:** The aim of the study is to determine the prevalence rates of ectoparasites (external parasites) seen in cattle in Muğla, Aydın and İzmir regions. **Methods:** For ectoparasite screening, a total of 1,353 cattle breeds from Muğla, Aydın and İzmir regions were included in the study. Ectoparasite samples were collected with the help of hand, forceps, comb and scalpel, brought to the laboratory in plastic containers containing 70% alcohol and examined under a microscope. **Results:** In the study, 191 (14.12%) of 1,353 cattle were infested with external parasites. Of the cattle, 138 (72.25%) ticks, 24 (12.57%) lice, 9 (4.71%) tabanid, 8 (4.19%) hippoboscid, 7 (3.66%) fleas and 5 (2.62%) mites were detected. Accordingly, 4 tick species in cattle *Hyalomma marginatum*, *Hyalomma excavatum*, *Hyalomma detritum*, *Rhipicephalus turanicus*, 4 lice species (*Bovicola bovis*, *Linognathus vituli*, *Solenopotes capillatus*, *Haematopinus quadripertusus*), 1 mite species (*Chorioptes bovis*), 2 flea species (*Ctenocephalides canis*, *Ctenocephalides felis*), 1 hippoboscid fly species (*Hippobosca equina*) ve 3 tabanid fly species (*Tabanus bromius*, *Tabanus exclusus*, *Philipomyia aprica*) diagnosed. **Conclusion:** The presence of external parasites varied depending on the frequency of antiparasitic use in cattle.

Peter, N., Dörge, D. D., Cunze, S., Schantz, A. V., Skaljic, A., Rueckert, S., & Klimpel, S. (2023). Raccoons contraband–The metazoan parasite fauna of free-ranging raccoons in central Europe. *International Journal for Parasitology: Parasites and Wildlife*, 20, 79-88.

Peterson, D. E., Dunnum, J. L., Liphart, S. W., Boegler, K. A., Eckerlin, R. P., Hoberg, E. P., ... & Cook, J. A. (2023). A Diverse Flea (Siphonaptera) Assemblage from the Small Mammals of Central New Mexico. *MANTER: Journal of Parasite Biodiversity* (ISSN 2470-8224) Occasional Papers, Number 31, April 10, 2023. doi: 10.32873/unl.dc.manter31

The geographical ranges of many mammals and their associated parasites are dynamic. Comprehensive documentation of these communities over time provides a foundation for interpreting how changing environmental conditions, driven by accelerating climate change, other anthropogenic disturbances, and natural events, may influence host-parasite interactions. Fleas (Order Siphonaptera) are obligate, hematophagous parasites of birds and mammals with medical interest because of their role in transmitting pathogens. From 2016 to 2019, we sampled the small mammal and associated flea communities in El Malpais National Conservation Area (El Malpais) in Cibola County, New Mexico. Among 898 mammalian specimens, 925 fleas representing 29 species were collected from 18 host species. *Pleochaetis exilis* was the most abundant flea species, composing 27% of the total fleas collected, whereas *Aetheca wagneri* was the most prevalent flea species, parasitizing 8% of the community sampled. Across a total of 284 hosts recorded with fleas, *A. wagneri*, *Malariaeus eremicus*, and *Peromyscopsylla hesperomys adelpha* parasitized the most host species (n = 6 each). *Onychomys leucogaster* (Wied-Neuwied, 1841), the northern grasshopper mouse, a rodent highly implicated in plague dynamics, was host for the highest number of flea species (n = 15), followed by *Peromyscus truei* (Shufeldt, 1885) (n = 10). Our aims are to (a) describe the flea-mammal assemblage of a central New Mexico site, creating a baseline for diversity against which changing patterns of association can be assessed over time; (b) identify previously unrecognized host associations; and (c) examine infestation parameters, including the relationships of flea prevalence and mean abundance to host sex, host abundance, and seasonality. As such, our study exemplifies the Documentation and Assessment phases of the DAMA protocol (Document, Assess, Monitor, Act), a central component of exploring distribution and diversity of complex pathogen-host communities across space and time that are



essential to a proactive understanding of emerging disease.

Prudhomme, J., Depaquit, J., Fite, J., Quillery, E., Bouhsira, E., & Liénard, E. (2023). Systematic review of hematophagous arthropods present in cattle in France. *Parasite*, 30.

Sadeddine, R., Righi, S., Daghri, K., Saidani, K., & Benakhla, A. (2023). A comparative study of the arthropod fauna in stork, barn swallow and common pigeon nests. *VETERINARSKA STANICA* 54 (5), 2023. | <https://doi.org/10.46419/vs.54.5.7>

Safarov, A., Nasreen, N., Akramova, F., Djabbarov, S., Mirzaeva, A., Esonboey, J., ... & Said, M. B. (2023). First report on ticks, mites, and other ectoparasites infesting carnivorous mammals in Uzbekistan. *Adv. Anim. Vet. Sci*, 11(8), 1297-1306.

Schantz, A. V., Dörge, D. D., Peter, N., & Klimpel, S. (2023). The hidden threat: Exploring the parasite burden and feeding habits of invasive raccoon dogs (*Nyctereutes procyonoides*) in central Europe. *International Journal for Parasitology: Parasites and Wildlife*, 22, 155-166.

Starikov, Vladimir P. and Egorov, Sergei V. and Vershinin, Evgeniy A. and Bernikov, Kirill A., 2023 04 48 507, Russian, Moscow, Rossiiskii Parazitologicheskii Zhurnal, (319–330), All-Russian K. I. Skryabin Scientific Research Institute of Helminthology, Fleas (Siphonaptera) of small mammals in the northern taiga of Western Siberia., (2023).

The purpose of the research is to identify the composition of fleas, and infection rates in small mammals in the northern taiga of Western Siberia; and to summarize information on fleas in the studied area. Materials and methods. In 2008, 2010, and 2016-2021, records of small mammals and their ectoparasites (fleas) were made in the northern taiga of Western Siberia. Parasitological survey was performed for 1363 specimens of insectivores and rodents of 13 species that were recorded using trap trenches, drift fences and trap-lines. A total of 1524 fleas of 18 species were recorded. Indices widely accepted in parasitology were used to assess quantitative indicators of fleas. Results and discussion. Based on the results of our own research and literature sources, small mammals in the northern taiga of Western Siberia were found to have specimens of 28 flea species. The flea fauna of the study area was represented by shrew fleas (*Corrodopsylla birulai*, *Palaeopsylla sorecis*), bird fleas (*Ceratophyllus gallinae*, *C. garei*), and specific species (*Ceratophyllus anisus*, *C. indages*, *C. sciurorum*, *Megabothris walkeri*, *Leptopsylla segnis*). *Peromyscopsylla silvatica*, *Corrodopsylla birulai*, *Megabothris rectangulatus*, *Amalaraeus penicilliger*, and *Amphipsylla sibirica* comprise the backbone of the flea fauna of small mammals in the northern taiga of Western Siberia.

Teixeira, R. H. F., da Costa, A. L. M., Gonzaga, C. R. R., Matos, F. N., de Medeiros, M. A., Leandro, S. D. F. S., ... & Barros, L. A. (2023). Flea parasitism in wild mammals in the metropolitan region of Sorocaba, São Paulo, Brazil. *Brazilian Journal of Veterinary Medicine*, 45.

Ukwueze, C. S., Unigwe, R. C., Uzoka, U. H., Onyeabor, A., & Olorok, W. N. (2023). PREVALENCE OF ECTOPARASITES OF DOGS IN Umuahia North and Ikwuano Local Government Areas of Abia State, Nigeria. *Animal Research International*, 20(3).

Vargas Charry, M. F. (2023). Pulgas con “casco” (Siphonaptera: Stephanocircidae) asociadas a mamíferos en Colombia: una revisión histórica, nuevos registros y establecimiento de nuevas asociaciones. <https://repositorio.ucaldas.edu.co/handle/ucaldas/19549>

The fleas of the Stephanocircidae family are known as "helmet fleas" due to a distinctive morphological modification in their head that resembles a helmet. This adaptation allows them to navigate through the fur of their hosts more easily and includes an additional ctenidium that provides them with enhanced anchorage. Stephanocircidae is divided into two subfamilies: Stephanocircinae, distributed in Australia, and Craneopsyllinae, distributed in South America. Craneopsyllinae is composed of seven genera, three of which, Cleopsylla, Plocopsylla, and Sphinctopsylla, have isolated records in Colombia. Studies on these fleas in the country and South America are scarce and fragmented. Therefore, to consolidate knowledge and understand the ecology of helmet fleas in Colombia, a systematic review of the records of Cleopsylla, Plocopsylla, and Sphinctopsylla, as well as the documented associations with their hosts in South America between 1911 and 2023, was conducted. Additionally, noteworthy new records and associations with bats in Colombia are presented. Based on this information, the distribution was evaluated, and a parasite-host interaction network for South America was constructed. In this study, the first records of Cleopsylla townsendi and Plocopsylla scotinomi in Colombia are presented. Furthermore, the distribution of seven species is expanded to the departments of Arauca, Boyacá, and Caldas, and a new association between Cleopsylla townsendi and Plocopsylla thor with the bat Myotis handleyi is recorded in the country. Distribution patterns and interactions revealed a close association between Plocopsylla with Cleopsylla and Sphinctopsylla, and apparent competition is observed between the species of the latter two genera. Finally, the flea-host interaction network reflects that these fleas exhibit a low degree of specialization, meaning their behavior is primarily generalist. However, their preferred hosts include rodents and other Andean-distributed mammals, which explains their distribution.

YILDIZ, Ö., & Onur, K. Ö. S. E. (2023). Investigation of ectoparasites in budgerigar and canaries in Burdur city of Turkey. Veterinary Journal of Mehmet Akif Ersoy University, 8(2), 89-94.

Zhang, L., Wei, X. H., Chang, N., Li, H. Y., Si, X. Y., Bai, X. F., ... & Liu, Q. Y. (2023). Composition of parasitic fleas on rodents and spatiotemporal distribution of flea index in xilin gol league, Inner Mongolia Autonomous Region, China, 2012-2021. Zhongguo Meijie Shengwuxue ji Kongzhi Zazhi = Chinese Journal of Vector Biology and Control, 16 February 2023, Vol. 34, No. 1, 82-90 ref. 24

To investigate the species composition of rodents and ectoparasitic fleas on rodents as well as the spatiotemporal distribution and changes of the flea index of dominant rodents in Xilin Gol League, Inner Mongolia Autonomous Region, China, and to provide a basis for plague risk assessment and its prevention and control decisions-making. Methods: With the monitoring data of 12 plague surveillance points in Xilin Gol League from 2012 to 2021, we used descriptive epidemiological methods to analyze the temporal and spatial distribution characteristics of main rodents and their flea indexes. Results: From 2012 to 2021, a total of 50 551 rodents were captured in Xilin Gol League, involving 18 species, 13 genera, and 4 families. Among them, 19 958 rodents carried fleas, with an average flea infestation rate of 39.48%. A total of 83 147 fleas were collected, belonging to 35 species/subspecies, 14 genera, and 5 families, with an overall flea index of 1.64. Meriones unguiculatus, Sperophilus dauricus, and Lasiopodomys brandtii were the dominant rodent species. Citellophilus tesquorum mongolicus, Xenopsylla conformis conformis, Frontopsylla luculenta, and Nosopsyllus laeviceps kuzenkovi were the main parasitic fleas. M. unguiculatus carried more diverse flea species on the body surface than L. brandtii and S. dauricus. The mixed flea index ( $S=-25.000$ ,  $P=0.032$ ) and S. dauricus flea index ( $S=-23.000$ ,  $P=0.049$ ) showed a downward trend in Xilin Gol League during the ten years. The M. unguiculatus flea index and L. brandtii flea index had no significant changes. The S. dauricus flea index was significantly higher than the L. brandtii flea index ( $H=15.000$ ,  $P < 0.001$ ) and the M. unguiculatus flea index ( $H=-15.000$ ,  $P < 0.001$ ). There was no significant difference between the L. brandtii and M. unguiculatus flea indexes ( $H < 0.001$ ,  $P=1.000$ ). The mixed flea index peaked in July at

4.47. The *M. unguiculatus* flea index was significantly different between banners/counties of Xilin Gol League ( $H=52.579$ ,  $P < 0.001$ ), which was highest in Sonid Left and Xianghuang banners. The *S. dauricus* flea index was significantly different between banners/counties of Xilin Gol League ( $H=73.253$ ,  $P < 0.001$ ), which was highest in West Ujimqin and Zhengxiangbai banners. The *L. brandtii* flea index showed no significant difference between banners/counties of Xilin Gol League ( $H=5.864$ ,  $P=0.118$ ). Conclusion: The three types of plague foci in Xilin Gol League overlap. *Meriones unguiculatus* carries a diversity of fleas, with a high risk of transmitting plague bacteria. Surveillance points should comprehensively monitor plague epizootics to prevent plague from resurging among animals and expanding to human beings.

### Physiology

Ajayi, O. M., Wynne, N. E., Chen, S. C., Vinauger, C., & Benoit, J. B. (2023). Sleep: An essential and understudied process in the biology of blood-feeding arthropods. *Integrative and Comparative Biology*, 63(3), 530-547.

Ceesay, M. (2023). A bioinformatics approach for evaluating evolutionary convergence of gene family size in hematophagous insects. Theses, Dissertations and Culminating Projects. 1202. <https://digitalcommons.montclair.edu/etd/1202>

The act of blood-feeding can be nutritionally rewarding for blood-feeding arthropods. However, blood digestion can release pro-oxidant molecules such as heme and iron at potentially harmful levels. If left uncontrolled, this heme/iron can cause oxidative damage and eventually cell death. This has led to the evolution of various adaptations that protect blood-feeding arthropods against iron- and heme-associated damage. Here I postulate that the signature of this adaptation can be observed in patterns of gene family size. To test this hypothesis, I explore convergent evolutionary expansions and contractions of gene families in distinct lineages of hematophagous insects. Specifically, I compare the gene content present in available genomes from blood-feeding and non-blood feeding arthropods (including outgroup taxa in the Lepidoptera [moths & butterflies]), to identify possible changes in gene family size in the blood-feeding taxa. Of the 206 heme/iron-associated genes identified from the model insect, *Drosophila melanogaster*, five were overrepresented (potentially duplicated) in the blood-feeding taxa: spook (*cyp307A1*), spookier (*cyp307A2*), cytochrome P450 12e1 (*cyp12e1*), hormone receptor and 51 (*Hr51*), NADH dehydrogenase (ubiquinone) B16.6 subunit (ND-B16.6), and seven were underrepresented (potentially lost). However, when only Dipteran (fly) and Siphonaptera (flea) genomes were included in the analysis, just one iron gene and one heme gene (NADH dehydrogenase (ubiquinone) PDSW subunit (ND-PDSW)) were overrepresented in the blood-feeding taxa. Interestingly, the expanded cytochrome genes are known detoxifiers of many compounds, including heme and iron. More broadly, the analytical approach I employed here could be used to evaluate functional convergence for other phenotypic traits, conditional on the availability of annotated genomic data.

Dou, X., & Jurenka, R. (2023). Pheromone biosynthesis activating neuropeptide family in insects: a review. *Frontiers in Endocrinology*, 14, 1274750.

"WFGPRLamide [neuro]peptide produced by capa is found in almost all insects, with a few exceptions (11). It was not found in three species of Siphonaptera and only in the Symphyta of Hymenoptera."

Fitzpatrick, L. L. J., Ligabue-Braun, R., & Nekar, K. A. I. (2023). Slowly Making Sense: A Review of

the Two-Step Venom System within Slow (*Nycticebus* spp.) and Pygmy Lorises (*Xanthonycticebus* spp.) [primate]. *Toxins*, 15(9), 514.

Gau, J., Lynch, J., Aiello, B., Wold, E., Gravish, N., & Sponberg, S. (2023). Bridging two insect flight modes in evolution, physiology and robophysics. *Nature*, 622(7984), 767-774.

Goode, C. K., Woodrow, C., Harrison, S. L., Deeming, D. C., & Sutton, G. P. (2023). Control of high-speed jumps in muscle and spring actuated systems: a comparative study of take-off energetics in bush-crickets (*Mecopoda elongata*) and locusts (*Schistocerca gregaria*). *Journal of Comparative Physiology B*, 193(6), 597-605.

Guo, L., Tang, M., Luo, S., & Zhou, X. (2023). Screening and Functional Analyses of Novel Cecropins from Insect Transcriptome. *Insects*, 14(10), 794.

Antibiotic resistance is a significant and growing threat to global public health. However, antimicrobial peptides (AMPs) have shown promise as they exhibit a broad spectrum of antibacterial activities with low potential for resistance development. Insects, which inhabit a wide range of environments and are incredibly diverse, remain largely unexplored as a source of novel AMPs. To address this, we conducted a screening of the representative transcriptomes from the 1000 Insect Transcriptome Evolution (1KITE) dataset, focusing on the homologous reference genes of Cecropins, the first identified AMPs in insects known for its high efficiency. Our analysis identified 108 Cecropin genes from 105 insect transcriptomes, covering all major hexapod lineages. We validated the gene sequences and synthesized mature peptides for three identified Cecropin genes. Through minimal inhibition concentration and agar diffusion assays, we confirmed that these peptides exhibited antimicrobial activities against Gram-negative bacteria. Similar to the known Cecropin, the three Cecropins adopted an alpha-helical conformation in membrane-like environments, efficiently disrupting bacterial membranes through permeabilization. Importantly, none of the three Cecropins demonstrated cytotoxicity in erythrocyte hemolysis tests, suggesting their safety in real-world applications. Overall, this newly developed methodology provides a high-throughput bioinformatic pipeline for the discovery of AMP, taking advantage of the expanding genomic resources available for diverse organisms.

["It is worth noting that cecropin was not identified previously, nor was the antimicrobial activity validated from the Siphonaptera and Mecoptera"].

Holst, J. D., Murphy, L. G., Gorman, M. J., & Ragan, E. J. (2023). Comparison of insect and human cytochrome b561 proteins: Insights into candidate ferric reductases in insects. *Plos one*, 18(12), e0291564.

King, B. H., & Gunathunga, P. B. (2023). Gustation across the class Insecta: body locations. *Annals of the Entomological Society of America*, 116(2), 76-82.

This review summarizes which body parts have taste function in which insect taxa. Evidence of taste by mouthparts, antennae, and tarsi is widespread. Mouthparts that commonly have taste function are the labium, including the labella and labial palps, the maxillae, including the galeae and maxillary palps, the inner surface of the labrum or clypeolabrum of chewers, and inside the precibarium/cibarium of hemipterans, which have piercing-sucking mouthparts. Tasting with mandibles has not been found, and tasting with the hypopharynx is seldom reported. Use of the antennae appears uncommon among fly species, but common among species of lepidopterans, hymenopterans, beetles, and bugs. Although

tasting with legs, especially tarsi, is reported mostly for fly and lepidopteran species, there is also evidence of it for multiple species of beetles, grasshoppers, and hemipterans, and one species of a roach, an ant, and a bee. Ovipositor taste function has been supported for some species of flies, lepidopterans, hymenopterans, orthopterans, and odonates. Taste by wings has been much less studied, but has been documented in a few fly species. Taste remains unstudied for any species or any body parts of Archaeognatha, Dermaptera, Mantodea, Mecoptera, Phasmatodea, Megaloptera, Neuroptera, Phthiraptera, Psocoptera, Siphonaptera, as well as Raphidioptera, Strepsiptera, Embioptera, Notoptera, and Zoraptera. Across holometabolous insects, larvae have not often been examined, the exception being some species of lepidopterans, flies, and beetles. Taste studies of antenna and legs are uncommon for even lepidopteran and beetle larvae.

Lamattina, D., & Salomón, O. D. (2023). *Triatoma infestans*, to be or not to be autogenic?. *Acta Tropica*, 237, 106727.

[No flea is known to develop eggs without blood meal. Editor]

Lu, S., Andersen, J. F., Bosio, C. F., Hinnebusch, B. J., & Ribeiro, J. M. (2023). Acid phosphatase-like proteins, a biogenic amine and leukotriene-binding salivary protein family from the flea *Xenopsylla cheopis*. *Communications Biology*, 6(1), 1280.

The salivary glands of hematophagous arthropods contain pharmacologically active molecules that interfere with host hemostasis and immune responses, favoring blood acquisition and pathogen transmission. Exploration of the salivary gland composition of the rat flea, *Xenopsylla cheopis*, revealed several abundant acid phosphatase-like proteins whose sequences lacked one or two of their presumed catalytic residues. In this study, we undertook a comprehensive characterization of the tree most abundant *X. cheopis* salivary acid phosphatase-like proteins. Our findings indicate that the three recombinant proteins lacked the anticipated catalytic activity and instead, displayed the ability to bind different biogenic amines and leukotrienes with high affinity. Moreover, X-ray crystallography data from the XcAP-1 complexed with serotonin revealed insights into their binding mechanisms.

Mathew, B., Aoyagi, K. L., & Fisher, M. A. (2023). Antibacterial activity of *Xenopsylla cheopis* attacins against *Yersinia pestis*. *BioRxiv*, 2023-08.

Palavecino, C. C., Fantozzi, M. C., Cuervo, P. F., Antoniazzi, L. R., Beldomenico, P. M., & Racca, A. L. (2023). Pro-and anti-inflammatory cytokine expression in wild *Akodon azarae*: individual variability due to host factors, seasonality and parasite infections. *Mammalian Biology*, 103(4), 405-417.

Sousa e Silva, R., Sousa, A. D., Vieira, J., & Vieira, C. P. (2023). The Josephin domain (JD) containing proteins are predicted to bind to the same interactors: Implications for spinocerebellar ataxia type 3 (SCA3) studies using *Drosophila melanogaster* mutants. *Frontiers in Molecular Neuroscience*, 16, 1140719.

Terra, W. R., Ferreira, C., & Silva, C. P. (2023). *Molecular Physiology and Evolution of Insect Digestive Systems* (Vol. 7). Springer Nature.

[Editor's note: Authors claim that fleas lack a peritrophic membrane because of midgut countercurrents and small size (Terra and Ferreira 2012). However, Chaika (1981) found one in flea larvae.]

## Evolution

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- Laslo, M., Just, J., & Angelini, D. R. (2023). Theme and variation in the evolution of insect sex determination. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*, 340(2), 162-181.
- Li, M., Kasan, K., Saha, Z., Yoon, Y., & Schmidt-Ott, U. (2023). Twenty-seven ZAD-ZNF genes of *Drosophila melanogaster* are orthologous to the embryo polarity determining mosquito gene *cucoid*. *Plos one*, 18(1), e0274716.
- Liu, P., Li, G., Zhao, N., Song, X., Wang, J., Shi, X., ... & Lu, L. (2023). Neutral forces and balancing selection interplay to shape the major histocompatibility complex spatial patterns in the striped Hamster in Inner Mongolia: suggestive of broad-scale local adaptation. *Genes*, 14(7), 1500.
- Puzakova, L. V., & Puzakov, M. V. (2023). Structure and Evolution of the AqE Gene in Insects. *Molecular Biology*, 57(1), 47-60.

## Ecology

- Baltrūnaitė, L., & Kitrytė, N. (2023). Seasonal dynamics and factors shaping ectoparasite communities of small mammals in Lithuania. In 10th Conference of the Scandinavian-Baltic Society for Parasitology (SBSP): June 5-7, 2023, Tartu, Estonia: book of abstracts. Ministry of rural affairs.
- Barriga-Carbajal, M. L., Vargas-Sandoval, M., & Mendoza, E. (2023). Deforestation increases the abundance of rodents and their ectoparasites in the Lacandon forest, Southern Mexico. *Revista de Biología Tropical*, 71(1).
- Caron-Lévesque, M., & Careau, V. (2023). Of mice, ticks, and fleas: host behaviour and co-occurring parasites. *Canadian Journal of Zoology*, 101(7), 510-521.

Host behaviour should be a key determinant of the acquisition of “sit-and-wait” and nidicolous ectoparasitic arthropods that ambush the host in their environment and/or nest. Here, we tested the association between parasitism and individual differences in exploratory and grooming behaviours in white-footed mice (*Peromyscus leucopus* (Rafinesque, 1818)), the primary host for the black-legged tick (*Ixodes scapularis* Say, 1821) in its larval stage. During 4333 captures of 1035 individual mice over five summers (2016–2020) in Ontario, Canada, the probability and intensity of tick parasitism were both significantly higher when the mouse was also parasitized by fleas, suggesting co-occurrence of these two parasites on host mice. Distance moved by mice in a novel environment was negatively and positively affected by tick and flea parasitism, respectively. Interestingly, there was a significant

“tick × flea” statistical interaction on distance moved, such that fleas were positively associated with activity/exploration only when ticks were absent. There was no relationship between grooming behaviour and parasite presence. This study suggests that co-occurring parasite species (ticks and fleas) may differentially affect their host's behaviour depending on the presence/absence of the other parasite on the host. Alternatively, host behaviour may differentially affect individual susceptibility to being infested with ticks, fleas, or both.

Davis, K. M. (2023). How Does Climate Change Influence Parasite Pressure on Birds? (Doctoral dissertation, The University of Utah).

Guarino, A. (2023). Size matters: Host body mass overshadows climate change in parasite prevalence among semi-arid ground squirrels (*Xerus inauris*). <http://hdl.handle.net/1993/37795>

Jakoniko, J., Massawe, A., Mwega, E., & Kessy, S. (2023). Associated factors contributing abundance of fleas on rodents in plague endemic area of Karatu district, northern Tanzania. Authorea Preprints not peer reviewed. TechRxiv. July 10, 2023. DOI: 10.22541/au.168845050.03946115/v2

Fleas are small wingless hematophagous insect that are frequently infesting on rodents and other small mammals while acting as reservoirs and vectors of many rodent borne zoonotic diseases including plague infectious disease which is threat to the public health in many part of the world including Tanzania. 291 rodents from nine species were captured with Sherman traps in farm land, peridomestic areas, bush land and forest buffer zone across wet and dry season in plague and non-plague foci villages. Captured rodents were anaesthetized and 190 fleas comprising four species were collected and morphologically identified with available dichotomous key. *Dinopsyllus lypus* were (46.32%), *Ctenophthalmus* spp (26.84%), *Xenopsylla brasiliensis* (16.32%) and *Xenopsylla cheopis* (10.53%). 38.42% of fleas were collected from *Mastomys natalensis*, 22.63% from *Lemniscomys striatus* and 18.42% from *Rattus rattus*. Highest flea infestation prevalence was found on *R. rattus* and was strongly associated with *X. cheopis* and *X. brasiliensis*. Specific flea index of *X. cheopis* on *R. rattus* was (01) in plague foci and (<0.5) in non-plague foci villages. Result of GLM final model indicated that flea abundance was significant influenced by rodent species ( $p < .001$ ), season ( $p = .031$ ), habitat type ( $p = .02$ ), rodent weight ( $p < .001$ ), rodent sex ( $p < .001$ ) and plague locations ( $p = .02$ ). There was significance difference in variation of flea abundance between rodent sexes ( $W = 9158.5$ ,  $p = .009$ ) and a weak positive correlation between rodent's weight and abundance of fleas ( $R = 0.17$ ,  $p < 0.05$ ). Despite that, specific flea index of *X. cheopis* on rats in both plague foci and non-plague foci villages were not indicating alarming condition that would require urgent control of fleas, still society should consistently adhere to rodent and fleas control methods in order to limit their interaction to the society especially in farm land and peridomestic areas where human activities are high.

Jackson, J. A., Bajer, A., Behnke-Borowczyk, J., Gilbert, F. S., Grzybek, M., Alsarraf, M., & Behnke, J. M. (2023). Remotely sensed localised primary production anomalies predict the burden and community structure of infection in long-term rodent datasets. *Global Change Biology*, 29(19), 5568-5581.

Krasnov, B. R., & Shenbrot, G. I. (2023). Evoregions of fleas and their small mammalian hosts: Do they coincide?. *Parasitology*, 150(11), 1031-1039.

Krasnov, B. R., Grabovsky, V. I., Khokhlova, I. S., Fernanda López Berrizbeitia, M., Matthee, S., Roll, U., ... & van der Mescht, L. (2023). Latitudinal distributions of the species richness, phylogenetic diversity, and functional diversity of fleas and their small mammalian hosts in four geographic quadrants. *Ecography*, e07129.

We studied latitudinal patterns in the species richness (SR), the phylogenetic diversity (PD), and the functional diversity (FD) of fleas and their mammalian hosts. We asked whether these patterns in either fleas, hosts, or both 1) conform to a classical latitudinal gradient; 2) vary geographically; and 3) differ between fleas and hosts. We also asked whether the patterns of PD and FD follow those of SR. We collected data on the latitudinal distribution of 1022 flea and 900 mammal species from literature sources and calculated the SR, PD, and FD of both groups in 1° latitude bands. Then, we used broken-stick regression models to analyse separately the latitudinal variation of 1) each diversity facet and 2) fleas and hosts in each geographic quadrant. The classical latitudinal gradient pattern was not found in either fleas or hosts across any facet of diversity or geographic quadrant, except for the PD of fleas in the southeastern quadrant and the FD of hosts in the southwestern quadrant. Latitudinal patterns of the SR, PD and FD of fleas and hosts differed substantially between geographic quadrants. Furthermore, the latitudinal distributions of flea and host SR were similar in three of four quadrants (except the northeastern quadrant), whereas the latitudinal distributions of flea and host PD were similar in the southwestern quadrant only. No similarity in flea versus host FD was revealed. The latitudinal patterns of flea and host PD and FD mostly did not follow those of their SR. We conclude that latitudinal gradients of species richness and phylogenetic and functional diversity appeared not to be universal phenomena. Instead, the latitudinal distributions of these diversity facets represent an interplay of ecological (current and past) and historical processes. For parasites, the processes acting on hosts add another layer of complexity underlying their latitudinal diversity patterns.

Kumari, T., Phogat, D., Phogat, J., Sonam, S., & Shukla, V. (2023). Dynamic Impacts of Weather Alteration and Urbanization on Entomofauna Diversity in Urban Ecosystem: A Study in the Rohtak District at Tilyar Lake and Maharshi Dayanand University. preprint not peer reviewed. <https://doi.org/10.21203/rs.3.rs-3410973/v1>

Langraf, V., Petrovičová, K., David, S., & Brygadyrenko, V. (2023). Comparison of spatial dispersion of epigeic fauna between alluvial forests in an agrarian and Dunajské luhy protected landscape area, southern Slovakia. *Cent. Eur. For. J.* 70 (2024) 3–10 DOI: 10.2478/forj-2023-0018

Mackenzie LS, Lambin X, Bryce E, et al. Patterns and drivers of vector-borne microparasites in a classic metapopulation. *Parasitology*. 2023;150(10):866-882. doi:10.1017/S0031182023000677

Sánchez-Loja, S., Donoso, D. A., & Páez-Vacas, M. I. (2023). Conspicuous and cryptic poison frogs are picky and prefer different meals in syntopy. *Evolutionary Ecology*, 1-13.

Trecker, R., Belthoff, J., & Iglesias Feliciano, P. (2023). An Unusual Host-Parasite Relationship: Factors Affecting the Rate at Which Burrowing Owls are Bugged by Fleas. *CONFERENCES > Idaho Conf Undergrad Res > 2023 > 138*

Zellner, P. N., & Brown, L. D. (2023). The effect of larval competition on adult life history traits and the immune response of adult cat fleas (*Ctenocephalides felis*). *Honors College Theses*. 908. <https://digitalcommons.georgiasouthern.edu/honors-theses/908>

In holometabolous insects, conditions during the larval stage can affect the adult stage. For instance, in mosquitoes, reduced larval food and high density can make adults more susceptible to infection. Fleas (Siphonaptera), like mosquitoes, go through complete metamorphosis and have adult stages that exclusively feed on the blood of vertebrates, including humans. However, flea larvae have chewing mandibles and rely on nutrients from dried fecal matter containing vertebrate blood proteins.



Their limited mobility and unique nutritional needs confine their development to specific areas, likely leading to frequent larval competition in natural conditions. Yet, the influence of larval density on the subsequent adult stage has not been explored. In our study using cat fleas (*Ctenocephalides felis*), we investigated the impact of larval competition on development to adulthood, adult body size, and adult immunity. We collected eggs to form three groups with different larval population densities ( $n = 50$ , 100, and 150), and maintained consistent nutritional conditions between all groups. We tracked development from eggs to pupae and adulthood, measured head and total body length of adults, and exposed eclosed adults to a blood meal with live bacteria (*Serratia marcescens*) for 24 hours to determine infection loads. Our findings indicate that low larval density negatively impacts the transition from pupa to adults, decreases adult size, and increased adult female susceptibility to infection. Overall, our study provides valuable insights into how larval density shapes the biology of resulting adult fleas.

### Pathology and Control

Zurita, A., Trujillo, I., & Cutillas, C. (2024). New records of pathogenic bacteria in different species of fleas collected from domestic and peridomestic animals in Spain. A potential zoonotic threat?. *Comparative Immunology, Microbiology and Infectious Diseases*, 102153.

Climate change is causing many vectors of infectious diseases to expand their geographic distribution as well as the pathogens they transmit are also conditioned by temperature for their multiplication. Within this context, it is worth highlighting the significant role that fleas can play as vectors of important pathogenic bacteria. For this purpose, our efforts focused on detecting and identifying a total of 9 bacterial genera (*Rickettsia* sp.; *Bartonella* sp.; *Yersinia* sp.; *Wolbachia* sp., *Mycobacterium* sp., *Leishmania* sp., *Borrelia* sp., *Francisella* sp. and *Coxiella* sp.) within fleas isolated from domestic and peridomestic animals in the southwestern region of Spain (Andalusia). Over a 19-months period, we obtained flea samples from dogs, cats and hedgehogs. A total of 812 fleas was collected for this study. Five different species were morphologically identified, including *C. felis*, *C. Canis*, *S. cuniculi*, *P. irritans*, and *A. erinacei*. *Wolbachia* sp. was detected in all five species identified in our study which a total prevalence of 86%. Within *Rickettsia* genus, two different species, *R. felis* and *R. asembonensis* were mainly identified in *C. felis* and *A. erinacei*, respectively. On the other hand, our results revealed a total of 131 fleas testing positive for the presence of *Bartonella* sp., representing a prevalence rate of 16% for this genus identifying two species *B. henselae* and *B. clarridgeiae*. Lastly, both *Y. pestis* and *L. infantum* were detected in DNA of *P. Irritans* and *C. felis*, respectively isolated from dogs. With these data we update the list of bacterial zoonotic agents found in fleas in Spain, emphasizing the need to continue conducting future experimental studies to assess and confirm the potential vectorial role of certain synanthropic fleas.

Zurita, A., Trujillo, I., García-Sánchez, Ángela María & Cutillas, C. (2024) Survey of flea infestation in cats and dogs in Western Andalusia, Spain: Seasonality and other risk factors for flea infestation. *Medical and Veterinary Entomology*, 1–8. Available from: <https://doi.org/10.1111/mve.12705>

This epidemiological survey aims to provide an update on the main flea species that parasitize domestic animals in the Western Andalusia assessing several ecological features that could be considered as possible risk factors for flea infestation. Over a 19-month period (June 2021 to January 2023), we obtained a total of 802 flea samples from 182 dogs (*Canis lupus familiaris*, Carnivora: Canidae, Linnaeus, 1758) and 78 cats (*Felis silvestris catus*, Carnivora: Felidae, Schreber, 1775). For each parasitized host, an epidemiological survey was completed, including the following information: geographical origin, age, sex, rural or urban habitat, type of animal's lifestyle (domestic or non-

domestic), health status, cohabiting or not with other animals and the total number of collected fleas. The most common species was *Ctenocephalides felis* (Siphonaptera: Pulicidae) (Bouché, 1835) with a total of 713 specimens, which accounted for 89% of the total fleas. The second most abundant species was *Pulex irritans* (Siphonaptera: Pulicidae) (Linnaeus, 1758) with a total of 46 collected fleas (6% of the total). The remaining species identified were *Archaeopsylla erinacei* (Siphonaptera: Pulicidae) (Bouché, 1835) (25 specimens), *Spilopsyllus cuniculi* (Siphonaptera: Pulicidae) (Dale, 1878) (12 specimens) and *Ctenocephalides canis* (Siphonaptera: Pulicidae) (Curtis, 1826) (6 specimens), which accounted for 3%, 1% and 1%, respectively, of the total fleas collected. The months with the highest number of collected fleas were, in ascending order, May 2022, September 2021 and July 2021. Dogs had a greater diversity of flea species, and flea sex ratios were female biased in all identified species and among all studied hosts. Finally, we identified some potential host risk factors that promoted higher flea intensities, such as living in rural areas, or presenting other pathologies.

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[Ctenocephalides, Dinopayllus, Hystrihopsylla, Leptopsylla, Xenopsylla]

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## Fleas in Art and History

Book Review  
*House of Trelawney*  
(2020)

by Hannah Rothschild, Bloomsbury Publ., London.

It's a fictional novel, but one of the characters is a "flea-trapper" and great aunt. She resembles Dame Miriam Rothschild, one of the great entomologists of the 20th century. The Trelawney family live on an ancient estate that sprawls along 40 miles of wooded riverside in the English countryside.

The great aunt lives away from the main house among the woods in a derelict cottage with microscopes, computers, journals, and tens of thousands of preserved flea specimens, at least until she's run out of hearth and home by her pet mice that served before escaping as hosts for her fleas. She frequently searches for and examines carefully carcasses in the woods and on roads to collect more fleas.

The great aunt wins world awards and makes important contributions to agriculture, conservation, and science. The reader learns that the British protect their bats better than the U.S. For the "flea-trapper", bats, rodents, and fleas are neither trivial nor hobby.

A whimsical and humorous portrait of a family, the novel perspective on the great aunt rounds out our understanding of eccentrics and geniuses. It could make an interesting movie.

– R.L. Bossard, Editor

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Deans, A. R., & Porturas, L. (2023). Diversity and complexity of arthropod references in haiku.  
<https://doi.org/10.31219/osf.io/mnkgb>

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"... with the popular title "Nature's Mysteries Disclosed", van Leeuwenhoek described and depicted the development of a "flea", from the egg via larval stages to an adult specimen, in remarkable detail."

Kutschera, U. (2023)  
Antonie van Leeuwenhoek (1632–1723): Master of Fleas and Father of Microbiology  
Microorganisms, 11(8), 1994

[Editor's note: Leeuwenhoek argued in a series of letters that his observations of the life cycle stages of the flea contradicted the view held by ancient biologists like Aristotle that fleas were spontaneously generated from dust. The flea's immature stages showed fleas were reproduced like larger animals.]

On the next page

Portrait of Anthonie van Leeuwenhoek (1632-1723), circa 1680  
by Jan Verkolje  
(baptized 9 February 1650, Amsterdam, Netherlands – buried 8 May 1693, Delft)  
(oil on canvas, height: 56 cm x width: 47.5 cm)





## Obituaries

### Dubinina Elena Vsevolodovna

(12.06. 1937-27.12. 2022)

Mironov, S. V., & Medvedev, S. G. (2023). In memoriam. Dubinina Elena Vsevolodovna (12.06. 1937-27.12. 2022). *Parazitologîa*, 57(1), 84-85.

### Ralph Peter Eckerlin

(New York City, 10 Feb. 1938 – Annandale, Virginia 9 Feb. 2024)

Obituary planned for *Journal of Parasitology*.

ASTMH Blog (/blog)>>February 2024 (/blog/february-2024)>>In Memoriam: Ralph Peter Eckerlin, PhD.

<https://www.astmh.org/blog/february-2024/in-memoriam-ralph-peter-eckerlin,-phd>

## THESES

Ph.D. Studies on the life cycle of *Strongyloides robustus* Chandler, 1942, and a survey of the helminths of Connecticut sciurids.

M.S. Life cycle studies on *Platynosomum fastosum* Kossack, 1910 (Trematoda: Dicrocoeliidae) in south Florida.

## PUBLICATIONS

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### Lice

*Polyplax guatemalensis* Durden & Eckerlin, 2001

### Helminths

*Eimeria hondurensis* Duszynski, Eckerlin & McCarthy, 2003

*Anoplocephaloides bulmeri* Haukisalmi & Eckerlin, 2009

*Arostrilepis gardneri* Makarikov, Galbreath, Eckerlin & Hoberg, 2020

*Arostrilepis insperata* Makarikov, Galbreath, Eckerlin & Hoberg, 2020

### Fleas

*Jellisonia painteri* Hastriter & Eckerlin, 2003

*Hystrihopsylla guatemalensis* Lewis & Eckerlin, 2004

*Catallagia appalachiensis* Eckerlin & Durden, 2023

*Jellisonia eckerlini* Hastriter, 2004

### Shrews

*Cryptotis eckerlini* Goodman, 2019 (“Ralph’s Broad-Clawed Shrew”)

